

Low-cost IOT based cold chain monitoring system for vaccine transport

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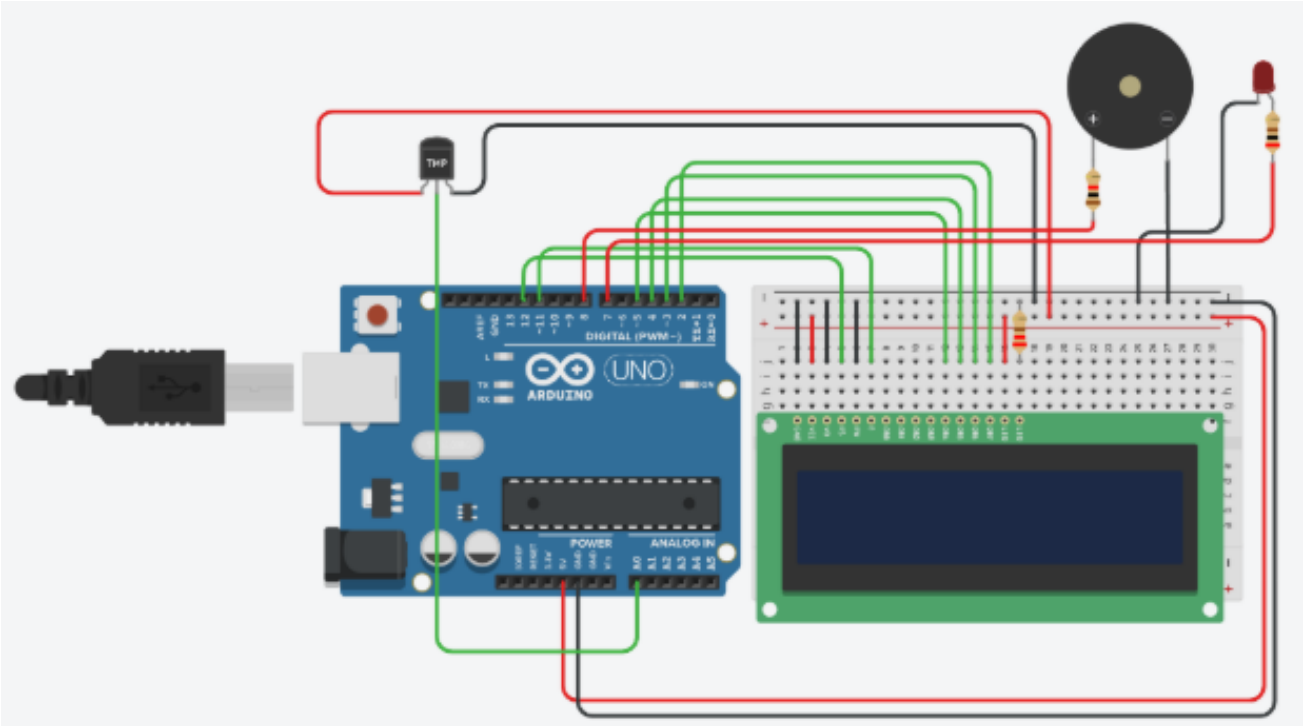


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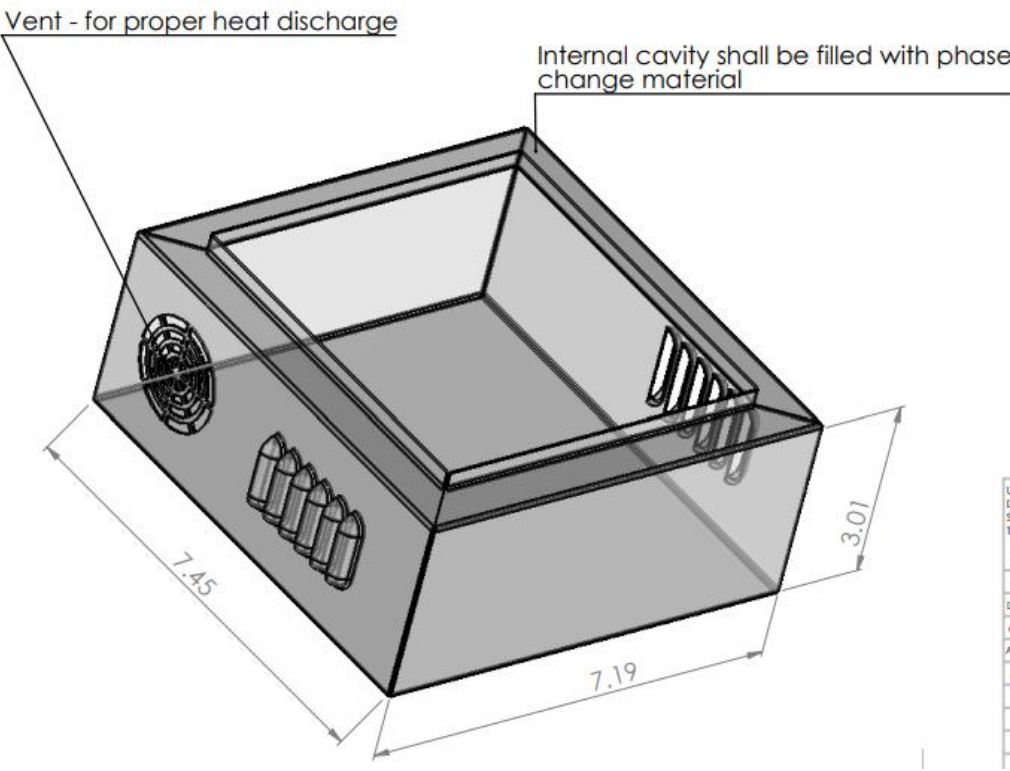
OVERVIEW

•The Cold Chain Monitoring System is an IoT-based project designed to track and maintain the temperature and location of perishable goods during storage and transport. It uses sensors and communication modules to provide real-time monitoring, alerts, and data logging, ensuring the integrity of temperature-sensitive products like vaccines and food.

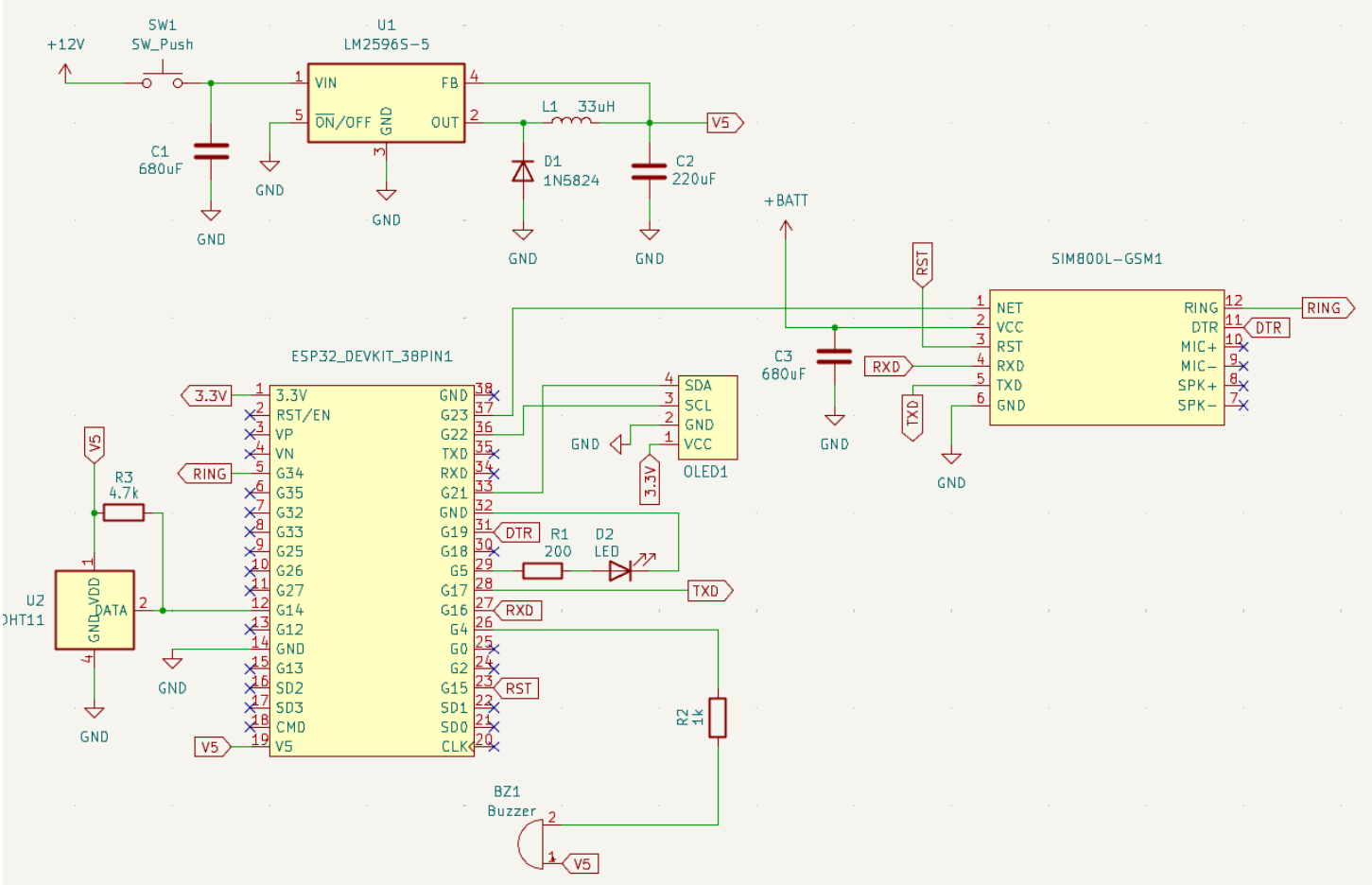
SIMULATION DIAGRAM



CAD MODEL



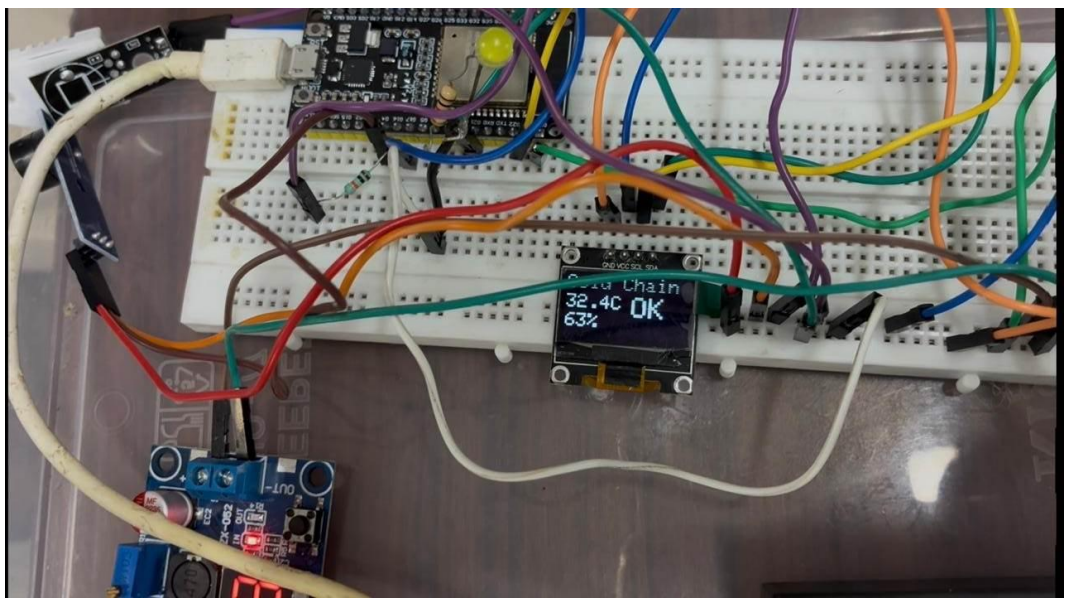
SCHEMATICS - PCB



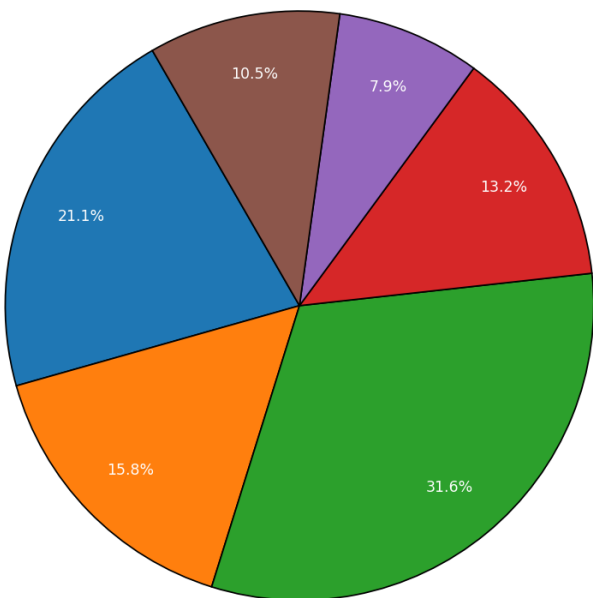
DESIGN CONSIDERATIONS

Environmental Parameters:
Temperature Range: -40°C to $+85^{\circ}\text{C}$
Humidity Monitoring: 0-100% RH
Accuracy: $\pm 0.5^{\circ}\text{C}$ temperature, $\pm 2\%$ humidity
Response Time: < 30 seconds
Power Requirements:
ESP32 System: 5V, 500mA
Peltier Module: 12V, 5A (60W)
PCM: No power required
Battery Backup: 8-12 hours operation
Communication:
WiFi 802.11 b/g/n
TCP/IP for LabVIEW integration
GSM/SMS for remote notifications
RFID 13.56 MHz

EXPERIMENTAL RESULTS

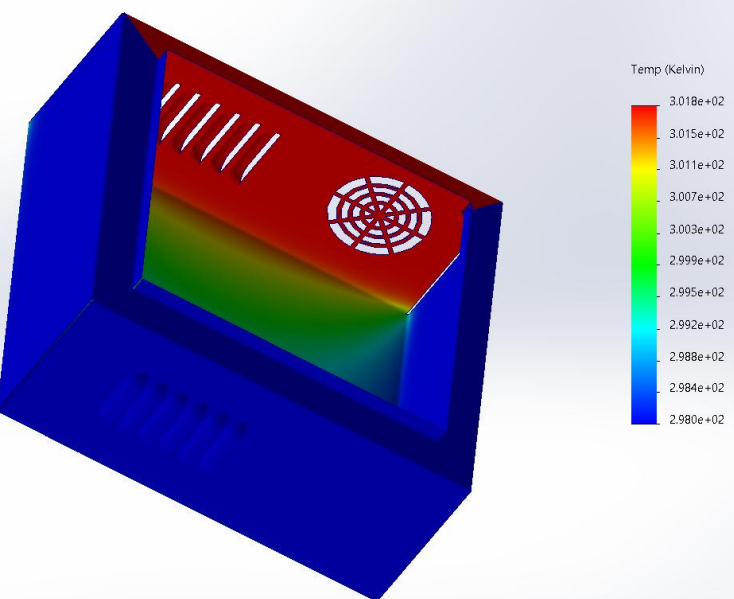
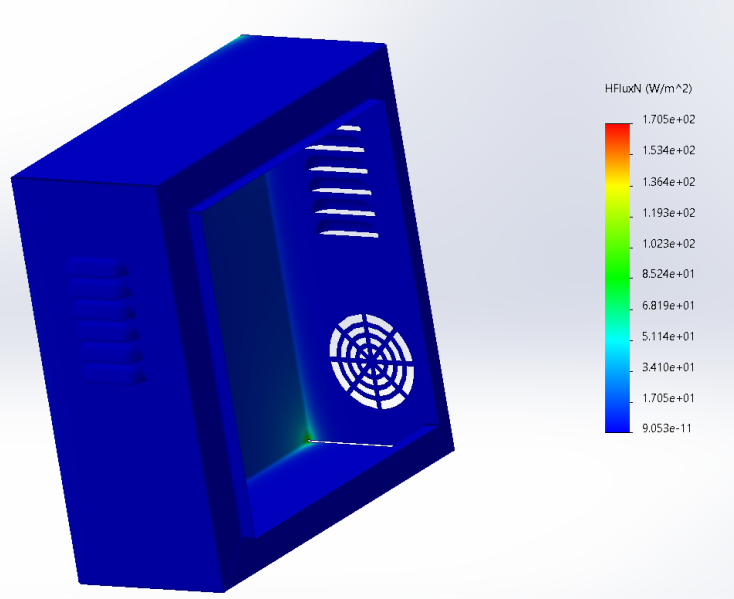


Cost Distribution of Cold Chain Monitoring Box



Components

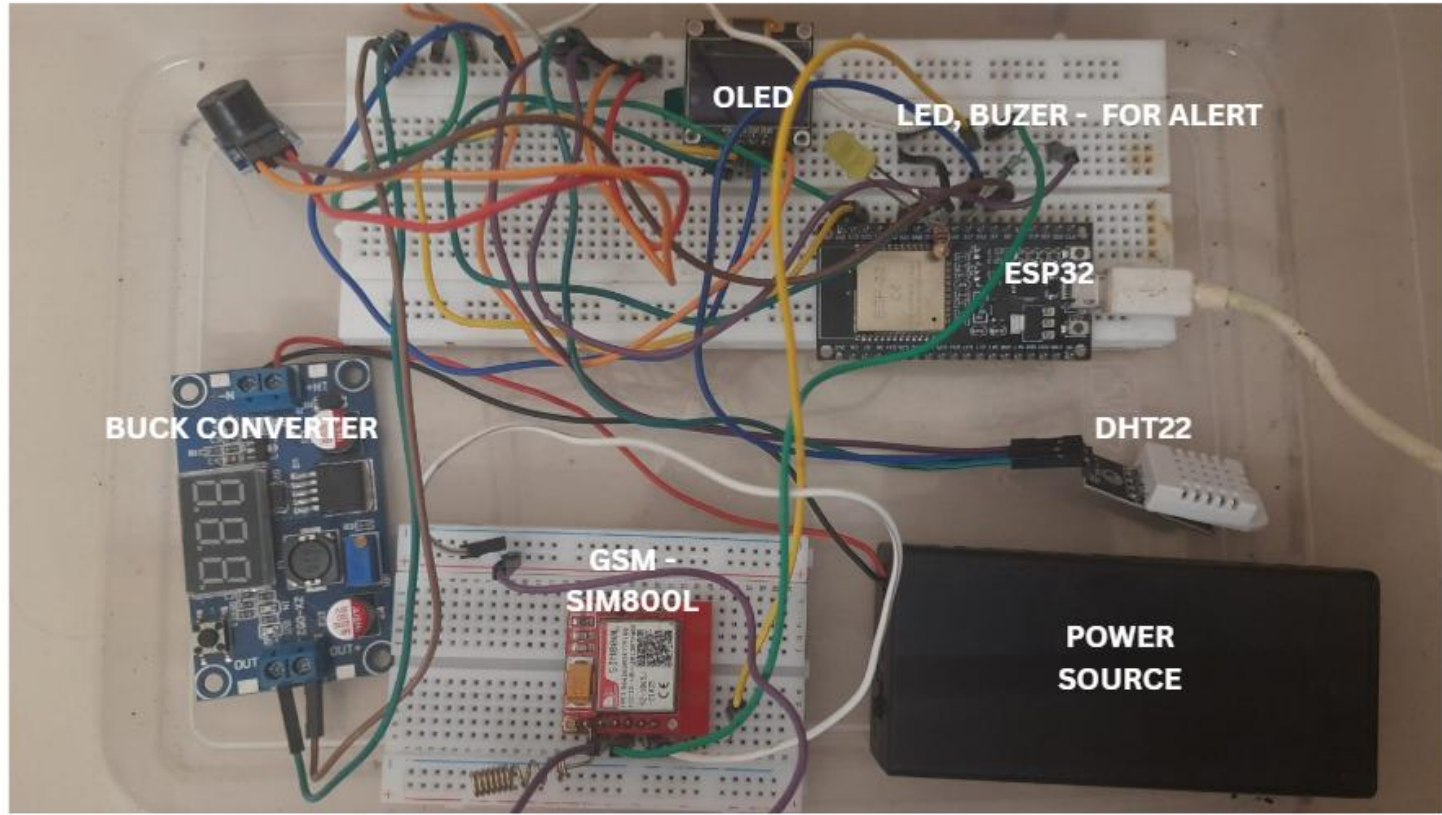
Enclosure (Plastic housing): ₹400
Thermal Management (PCM + Insulation): ₹300
Electronics (ESP32, SIM800L, PCB, sensors): ₹600
Power System (Battery, charger, converter): ₹230
Assembly + Misc.: ₹150
Testing & Packaging: ₹200



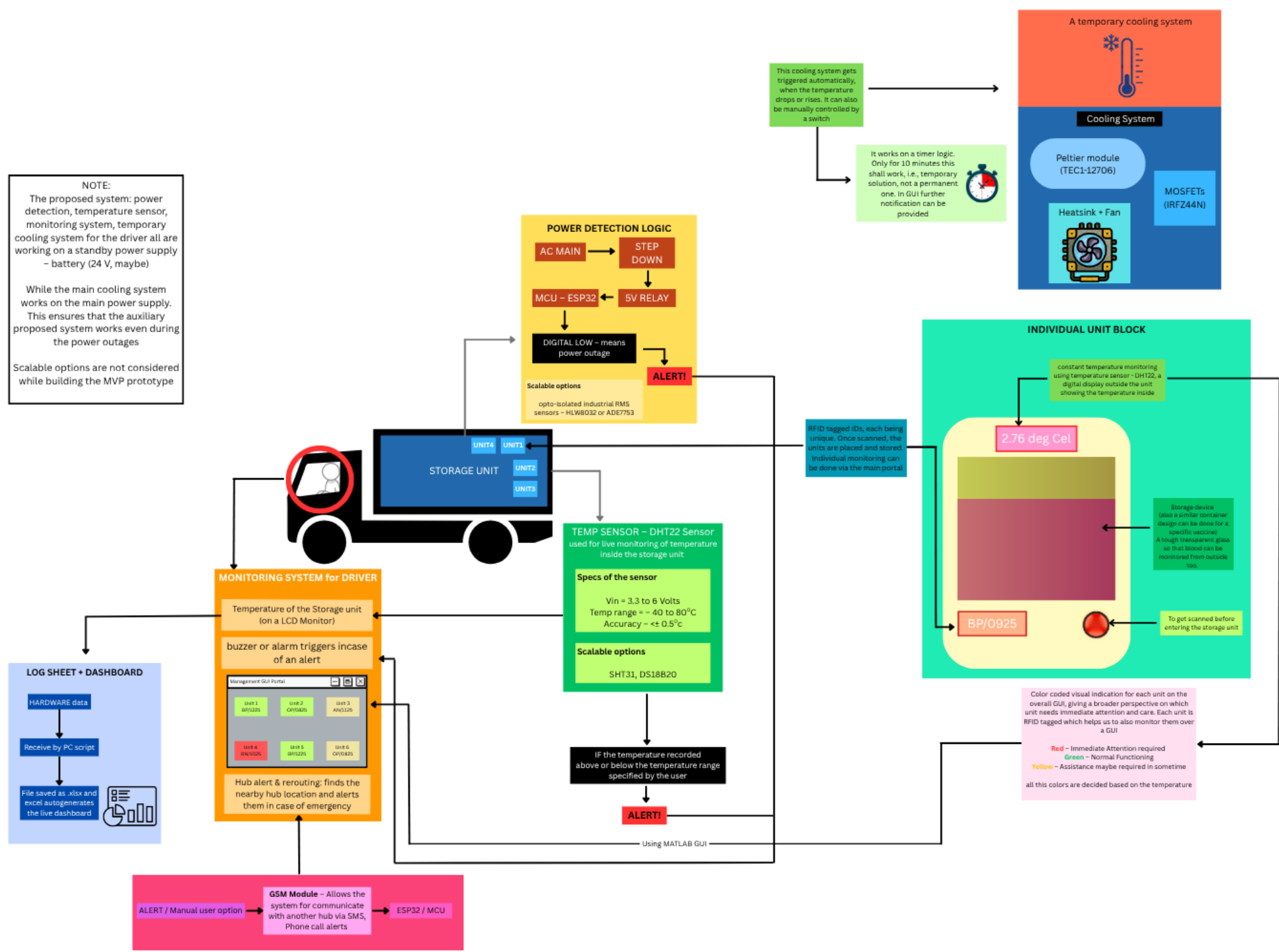
INFERENCES

- The project demonstrated a fully functional IoT-enabled Cold Chain Monitoring System capable of detecting and alerting temperature variations instantly.
- Analytical and hardware results verified its accuracy, stability and, reliability.
- By combining low-cost components with efficient real-time communication, the system ensures product integrity, reduces losses, and improves supply chain transparency. The project serves as a practical and scalable step toward smart logistics and digital cold chain management.

CIRCUIT PHOTOGRAPH



FLOW CHART



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